Multiclass Brain Tumor Segmentation in MRI Scans: A Comparative Study of 3D U-Net, Attention U-Net, and Swin UNETR

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Abstract—One major challenge in analyzing brain scans comes from separating tumors within MRI data, mainly due to their uneven shapes, mixed internal structure, nevertheless blending into surrounding areas that appear nearly identical. This study examines performance across three 3D deep networks built using the BraTS 2021 collection: a fundamental 3D U-Net architecture, an adaptation enhanced with attention mechanisms, alongside a transformer-based variant called Swin UNETR. Each network undergoes full training over 100 cycles via a blend of Dice and cross-entropy objectives, later tested on accepted regional divisions - entire growth, central mass, plus enhancing portion - with evaluation relying upon Dice values and HD95 metrics. Findings indicate the model incorporating attention delivers top Dice outcomes (WT: 0.9352, TC: 0.9077, ET: 0.8649), along with reduced HD95 measures for whole lesion and core sections, implying learned focus zones contribute stable improvements compared to basic skip connections. Although the Swin UNETR underperformed relative to potential capability, limited setup conditions likely played a role. With Grad-CAM visuals, focus shifts clearly toward Attention U-Net's activation patterns - tight, precise, aligned with anatomy. Guidance emerges quietly: when segmenting brain tumors in clinics, this model stands apart.

Index Terms—Brain tumor segmentation, BraTS, 3D U-Net, Attention U-Net, Swin UNETR, Dice coefficient, Hausdorff distance, Grad-CAM, deep learning, medical image analysis.

1 Introduction

High -grade brain tumors, especially gliomas, rank among the deadliest forms of cancer; survival time for those with glioblastoma averages just 14 to 16 months even when treated intensively. Defining distinct areas within the tumor - such as the dead center, surrounding swelling, and the outer active edge - is essential when preparing surgery, setting radiation targets, or tracking changes over time. In past practice, trained radiologists performed these evaluations, a process that takes considerable time, demands significant resources, and often shows variation between specialists. [1]. Accurate delineation of tumor sub-regions—the necrotic core, peritumoral edema, and enhancing rim—is fundamental for surgical planning, radiotherapy target definition, and longitudinal response assessment. Traditionally this has been done by expert radiologists, which is slow, expensive, and prone to inter-observer disagreement [2].

One type of MRI combines several types of imaging to show different aspects of tumors. While T1ce shows areas where the tumor enhances, T2/FLAIR displays swelling around it; meanwhile, native T1 aids in locating structures. Despite these benefits, combining all four scans presents difficulty - especially when predicting individual voxels across vast 3D spaces. Such tasks must manage uneven data distribution alongside diverse growth patterns.

The Brain Tumor Segmentation (BraTS) benchmark [2] Progress here has continued primarily because of its influence ever since 2012. It provides standardized, skull-stripped, co-registered volumes with expert consensus annotations for three nested sub-regions: Whole Tumor (WT), Tumor Core (TC), and Enhancing Tumor (ET). Performance is reported using Dice Similarity Coefficient (DSC) and 95th-percentile Hausdorff Distance (HD95).

In this project, we implement and compare three deep learning architectures on BraTS 2021: (1) a baseline **3D U-Net** [3], (2) an **Attention U-Net** [4] that uses soft gating to suppress irrelevant activations, and (3) a **Swin UNETR** [5] that replaces the convolutional encoder with a Swin Transformer for global context modeling. Our goal is not just to rank models, but to understand *why* one outperforms another—both through quantitative metrics and Grad-CAM saliency visualization.

The remainder of this paper is organized as follows. In section 2 we reviews relevant prior work. In section 3 we describes our dataset, preprocessing, architectures, and training protocol. Section 4 presents quantitative and qualitative results. Section 6 concludes with a discussion of limitations and future directions.

2 Related Work

Over recent years, automatic brain tumor segmentation has changed fast. Instead of relying on manually designed traits

or older model types, researchers now use full-scale deep learning systems. One of the earliest attempts involved CNNs applied directly to MRI data. Pereira and team [6] led early efforts by testing convolutional networks on BraTS scans through 2D image chunks, using tiny 3 × 3 kernels. Though their method worked well back then, analyzing slices separately missed connections between neighboring layers. Later, Havaei et al. [7] built a network with two parallel streams handling different resolution levels, linking outputs step by step - this cut down computation speed without losing precision. Moving into volume-based analysis, 3D methods began taking shape. DeepMedic, developed by Kamnitsas et al. [8], tackled shortcomings in 2D methods through a dual-path 3D convolutional network handling entire volume chunks at varying scales - later sharpened using a Conditional Random Field to refine edges. Rather than fading out, the U-Net design from Ronneberger et al. [3] stuck around, its encoder-decoder shape supported by shortcut links guiding accurate spatial recovery during tissue division tasks. Myronenko [9] built on that base by weaving in constraints from a Variational Autoencoder, boosting performance enough to claim victory in BraTS 2018. Attention entered via Oktay et al. [4], who wove gated controls into the decoder path of U-Net; these let signals grow stronger only where needed for segmentation success. Small elements, such as vivid tumor zones, gain clarity since raw merging across features often brings interference - but focused scaling cuts through it. Systems began adjusting themselves more freely after that shift toward adaptive designs. A study by Isensee et al. [10] showed that an automatically optimized U-Net variant - called nnU-Net - beats numerous new network designs, highlighting how setup details can outweigh structural originality. Rather than stacking layers differently, some researchers turned to attention mechanisms. Wang et al. [11], for example, introduced TransBTS, blending Transformer modules into a 3D convolutional backbone to capture broader spatial patterns. Building on that idea, Hatamizadeh et al. [5] designed UNETR, which swaps the entire encoder part with a Vision Transformer. Performance gains emerged when Swin UNETR [5] swapped in a hierarchical version of the Transformer, boosting both speed and precision in tests on BraTS 2021 data. Unlike earlier studies, our approach trains all three systems using the same settings while adding Grad-CAM visualization to support numerical findings.

3 METHODS

3.1 Dataset

We use the **BraTS 2021 Task 1** dataset [2], which contains 1,251 multi-parametric MRI cases collected from multiple institutions. Each case includes four co-registered, skull-stripped sequences: T1-weighted (T1), contrast-enhanced T1 (T1ce), T2-weighted (T2), and T2 Fluid Attenuated Inversion Recovery (FLAIR). All volumes are resampled to an isotropic $1\,\mathrm{mm}^3$ resolution, aligned to the SRI24 atlas, and stored in NIfTI format with native dimensions of $240 \times 240 \times 155$ voxels.

Voxel-wise annotations define four tissue classes: background (label 0), necrotic tumor core (label 1), peritumoral edema (label 2), and enhancing tumor (label 4). These are combined into three clinically relevant overlapping subregions: Whole Tumor (WT = labels 1+2+4), Tumor Core (TC = labels 1+4), and Enhancing Tumor (ET = label 4 only). This hierarchical label structure reflects direct clinical utility—WT captures the full extent of abnormal signal visible on FLAIR, TC targets the surgically critical core, and ET identifies actively proliferating tissue most visible on T1ce post-contrast.

Class imbalance is a defining challenge of this dataset: background voxels account for over 98% of each volume, and ET is the rarest label, sometimes covering less than 1% of the total scan. This imbalance directly motivates our choice of a combined Dice and cross-entropy loss. We apply a stratified 80/10/10 split, yielding approximately 1,001 training, 125 validation, and 125 test subjects, ensuring that diverse tumor morphologies are represented proportionally across all splits.

3.2 Preprocessing

All preprocessing is implemented in PyTorch using the MONAI framework [2]. The pipeline is kept consistent across all three models to ensure fair comparison. The following steps are applied sequentially to each subject:

- **Intensity normalization:** Each MRI channel is independently normalized using Z-score statistics computed over non-zero (brain mask) voxels only, i.e., $\hat{x} = (x - \mu_{\text{brain}})/\sigma_{\text{brain}}$. Voxels outside the skull-strip mask are set to zero. This is done per-patient to account for scanner variability across institutions.
- **Spatial cropping:** Although the raw volumes are $240 \times 240 \times 155$, the majority of voxels are background. We compute a tight foreground bounding box from the union of all non-zero label voxels, add a small margin of 10 voxels on each side, and then crop the volume. The result is further resized to a fixed $128 \times 128 \times 128$ patch, which fits within GPU memory while retaining the full tumor region.
- **Label remapping:** The original BraTS label set (0, 1, 2, 4) is remapped to (0, 1, 2, 3) for compatibility with PyTorch's indexing, and three binary masks are derived for WT, TC, and ET to match the evaluation protocol.
- **Data augmentation (training only):** We apply a stochastic augmentation pipeline with random left-right, anterior-posterior, and superior-inferior flips (each with $p = 0.5$), random intensity scaling by a factor uniformly sampled from $[0.9, 1.1]$, random Gaussian noise ($\sigma \in [0, 0.1]$) with $p = 0.15$, and a random affine transform (rotation up to $15°$, translation up to 10 voxels) with $p = 0.5$. Augmentation is applied only during training and is not used at validation or test time.

3.3 Model Architectures

3.3.1 3D U-Net (Baseline)

Our baseline follows the encoder-decoder design of Ronneberger et al. [3] extended to 3D volumetric inputs. The encoder consists of four resolution stages, each containing two $3 \times 3 \times 3$ convolutions with instance normalization and leaky

ReLU activations, followed by $2 \times 2 \times 2$ strided convolution for downsampling. The number of feature channels doubles at each encoder stage, progressing from 32 to 64, 128, and 256. A bottleneck stage applies two more convolution blocks at the coarsest resolution.

The decoder mirrors the encoder symmetrically: at each stage, the feature map is upsampled via transposed convolution (stride 2), concatenated with the corresponding encoder skip connection, and passed through two convolution blocks. The skip connections are crucial for recovering fine spatial detail lost during downsampling, which is especially important for delineating small structures like the ET boundary. A final $1 \times 1 \times 1$ convolution followed by sigmoid activation produces four independent binary output maps, one per segmentation channel (NCR, ED, ET, and a combined whole-brain mask).

The total parameter count for the 3D U-Net is approximately 19.1 million.

3.3.2 Attention U-Net

The Attention U-Net [4] builds on the 3D U-Net baseline by inserting soft spatial attention gates at each decoder skip connection. Formally, at resolution level l, the attention gate takes two inputs: a gating signal g^l from the coarser decoder stage (carrying semantic context) and the encoder feature map x^l (carrying spatial detail). Both are projected to an intermediate dimension via $1 \times 1 \times 1$ convolutions, summed, passed through ReLU, and then projected to a scalar map via another $1 \times 1 \times 1$ convolution followed by sigmoid:

$$\alpha^l = \sigma\big(W_\psi^\top \cdot \mathrm{ReLU}\big(W_g^T g^l + W_x^T x^l + b_g\big) + b_\psi\big) \quad (1)$$

The scalar attention map $\alpha^l \in [0,1]^{H \times W \times D}$ is upsampled to match x^l and applied voxel-wise, producing $\hat{x}^l = \alpha^l \odot x^l$. High values of α indicate spatially relevant regions for segmentation, while low values suppress background noise before concatenation in the decoder. This gating operation adds minimal parameters (the projection layers are small) but provides a substantial regularization benefit by focusing gradient flow on diagnostically relevant regions. All other components—encoder depth, channel progression, decoder structure, and output head—are identical to the 3D U-Net baseline, making the comparison fair and the contribution of attention gates directly measurable.

The total parameter count for Attention U-Net is approximately 19.6 million, just 0.5 million more than the baseline.

3.3.3 Swin UNETR

Swin UNETR [5] departs from the CNN paradigm by replacing the encoder entirely with a hierarchical Swin Transformer. The input volume of size $128 \times 128 \times 128$ is first partitioned into non-overlapping $2 \times 2 \times 2$ patches, each linearly embedded into a 48-dimensional token. These tokens are processed through four Swin Transformer stages, each applying Shifted Window Multi-head Self-Attention (SW-MSA) to model both local and global spatial dependencies efficiently:

- **Stage 1:** Window-based self-attention at full spatial resolution (token grid $64 \times 64 \times 64$), 3 Transformer blocks, 48 channels.
- **Stage 2:** After patch merging (resolution halved), 48 channels, 6 Transformer blocks.
- **Stage 3:** Further merging, 96 channels, 6 blocks. Long-range context captured at this scale.
- **Stage 4:** Coarsest scale, 192 channels, 3 blocks. Global context dominates.

The shifting window strategy alternates between regular and shifted window partitions across consecutive Transformer blocks, enabling cross-window information exchange without the quadratic cost of full global attention. Feature maps from each stage are fed as skip connections into a CNN-based decoder that mirrors the U-Net topology. We initialize the encoder with publicly available self-supervised pretrained weights, obtained by pretraining on a large collection of unlabeled CT and MRI volumes using masked volume inpainting as the pretext task.

Swin UNETR has approximately 62.2 million parameters—roughly three times more than either U-Net variant—reflecting the significantly larger capacity of the Transformer backbone.

3.4 Training Protocol

All models are trained under identical conditions to ensure a fair comparison. Table 1 summarizes the key hyperparameters.

TABLE 1
Training Hyperparameters (identical for all models)

Hyperparameter	Value
Loss function	Dice + Binary Cross-Entropy
Optimizer	Adam (β_1=0.9, β_2=0.999)
Initial learning rate	1×10^{-4}
Weight decay	1×10^{-5}
LR scheduler	Cosine annealing
Minimum LR	1×10^{-6}
Batch size	2 per GPU
Epochs	100
Input patch size	$128 \times 128 \times 128$
Mixed precision	FP16 (AMP)
Hardware	NVIDIA A100 40 GB

Loss function. We use a combination of soft Dice loss and binary cross-entropy, averaged over all four output channels:

$$\mathcal{L} = \mathcal{L}_{\text{Dice}} + \mathcal{L}_{\text{BCE}} = \left(1 - \frac{2\sum p_i g_i}{\sum p_i + \sum g_i}\right) - \sum g_i \log p_i \quad (2)$$

where $p_i \in [0,1]$ is the predicted probability and $g_i \in \{0,1\}$ is the ground-truth label for voxel i. The Dice component directly optimizes the evaluation metric and handles class imbalance naturally by focusing on overlap rather than per-voxel accuracy. The BCE term stabilizes training in early epochs when predictions are near-uniform.

Checkpoint strategy. We monitor mean validation Dice (average over WT, TC, ET) after each epoch and save the model weights whenever this metric improves. Final evaluation is always performed with the best saved checkpoint, not the last epoch weights. This is particularly important

for Swin UNETR, whose loss continues to decrease slowly beyond epoch 80 while validation Dice may have already plateaued.

Mixed precision training. We use PyTorch's Automatic Mixed Precision (AMP) with FP16 computation and FP32 gradient accumulation. This reduces GPU memory consumption by approximately 40% compared to full FP32 training, allowing larger effective batch processing without sacrificing numerical stability.

3.5 Evaluation Metrics

We report two standard BraTS metrics for each of the three sub-regions, following the official evaluation protocol:

Dice Similarity Coefficient (DSC) measures volumetric overlap between the predicted mask P and ground-truth G:

$$\mathrm{DSC}(P, G) = \frac{2|P \cap G|}{|P| + |G|} \tag{3}$$

DSC ranges from 0 (no overlap) to 1 (perfect overlap). It is sensitive to the overall volume of the prediction but relatively robust to boundary placement errors for large structures like WT.

95th-Percentile Hausdorff Distance (HD95) measures boundary accuracy. It is defined as the 95th percentile of the set of surface-to-surface distances:

$$\mathrm{HD95}(P, G) = \mathrm{perc}_{95}\left(\max_{p \in \partial P} d(p, \partial G),\ \max_{g \in \partial G} d(g, \partial P)\right) \tag{4}$$

where $d(p, S)$ is the minimum Euclidean distance from point p to surface S. By taking the 95th percentile rather than the maximum, HD95 is robust to isolated outlier predictions—which are common in difficult cases where the model predicts a small isolated false-positive island far from the true boundary. Lower values indicate tighter boundary alignment, which matters for radiotherapy margin delineation.

Together, DSC and HD95 provide complementary views: a model can achieve high DSC through correct volume estimation while still having poorly-defined boundaries (high HD95), or vice versa.

3.6 Interpretability: Grad-CAM

To understand what spatial regions each model uses to make its predictions, we apply Gradient-weighted Class Activation Mapping (Grad-CAM) [14] to the final convolutional feature map of each architecture. The procedure is as follows: for each output channel $c \in \{\text{NCR, ED, ET}\}$, we compute the gradient of the predicted segmentation score S^c with respect to the activations A^k of the target feature map layer, then global average pool the gradients over the spatial dimensions to obtain a scalar weight w_k^c for each feature map k:

$$w_k^c = \frac{1}{Z} \sum_{i,j,d} \frac{\partial S^c}{\partial A_{ijd}^k} \tag{5}$$

The Grad-CAM saliency map is then formed as:

$$\mathrm{GradCAM}^c = \mathrm{ReLU}\left(\sum_k w_k^c A^k\right) \tag{6}$$

TABLE 2
Test Set Performance on BraTS 2021 (Mean ± reported values)

Model	**Dice Score ↑**			**HD95 (mm) ↓**		
	WT	TC	ET	WT	TC	ET
3D U-Net	0.9312	0.9013	0.8593	2.481	2.491	1.778
Attn U-Net	**0.9352**	**0.9077**	**0.8649**	**1.762**	**2.344**	2.298
Swin UNETR	0.9265	0.8799	0.8488	1.997	2.937	2.921

The ReLU ensures we only visualize features that positively contribute to the class score. The resulting 3D saliency map is normalized to $[0, 1]$ and the central axial slice is overlaid on the corresponding T1ce MRI slice using a jet colormap for visualization. Red regions indicate where the model is most sensitive for that particular output channel.

4 Experiments and Results

4.1 Quantitative Results

Table 2 The performance results for each model on the test set across the three BraTS sub-regions are summarized in Table 2. The bar charts showing a comparison between the Dice scores and HD95 values of each model can be found in Fig. 3.

Attention U-Net achieves the highest Dice scores across all three sub-regions and the lowest HD95 on WT and TC, making it the top-performing model in our study. The improvement over the 3D U-Net baseline is consistent: $+0.004$ on WT, $+0.006$ on TC, and $+0.006$ on ET Dice. More strikingly, it reduces HD95 on WT by ~ 0.72 mm and on TC by ~ 0.15 mm, which is clinically meaningful for treatment planning. The one region where U-Net narrowly outperforms Attention U-Net is ET HD95 (1.778 vs. 2.298 mm), which might reflect a few difficult outlier cases where the attention mechanism over-suppresses enhancing voxels.

3D U-Net performs solidly as a baseline. It achieves competitive Dice scores and notably the best HD95 on ET (1.778 mm), which was somewhat unexpected. Looking at the boxplots (Fig. 2), the ET Dice distribution for U-Net is slightly tighter around the median compared to Attention U-Net, suggesting fewer catastrophic failure cases in that sub-region.

Swin UNETR delivers the weakest results among the three, underperforming the 3D U-Net on Dice (WT: -0.005, TC: -0.021, ET: -0.011) and showing substantially higher HD95 values on TC and ET. This result, while perhaps counterintuitive given the model's architectural capacity, is consistent with observations in the literature that Transformer-based models typically require either large training sets or extensive pretraining to reach their full potential. In our setup, both 3D U-Net and Attention U-Net converge to a lower training loss (~ 0.20 final) compared to Swin UNETR (~ 0.30 final), as shown by the training curves in Fig. 1. The higher residual loss of Swin UNETR indicates it is not fully exploiting the training data within 100 epochs.

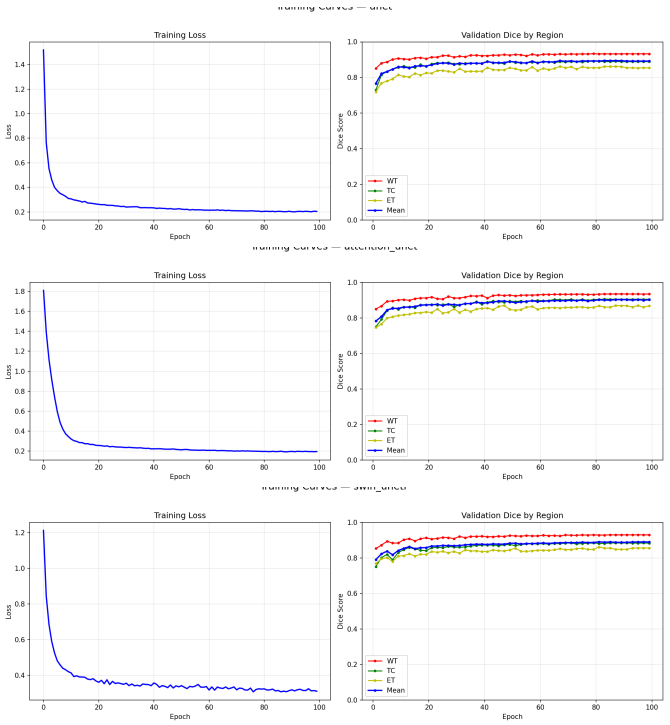

Fig. 1. Training loss (left) and validation Dice per region (right) over 100 epochs for 3D U-Net (top), Attention U-Net (middle), and Swin UNETR (bottom). Both U-Net variants converge smoothly and stabilize around epoch 40–50. Swin UNETR shows noisier convergence with a higher residual loss.

4.2 Training Curves

For all three models, the fast decline loss pattern is seen within the first 10 epochs, after which there is a slower descent. The 3D U-Net and Attention U-Net stabilize around 0.20 at epoch 50, maintaining their performance for the last 50 epochs. In terms of validation Dice per region, the same pattern can be observed for the test dataset as well. Here, the WT is the simplest to segment (red, top); TC comes second (green), while ET is the most unpredictable (yellow). The mean Dice value (blue) grows gradually and stabilizes around epoch 60.

As for Swin UNETR, it starts with a higher initial loss (1.22), even with pretrained weights, and stabilizes significantly later, reaching its plateau at 0.30. This means that the Transformer attention block might require more time for warming up on the given data size.

4.3 Dice Score Distributions

From Fig. 2, it can be seen that all models have median Dice values consistently very high (greater than 0.93 for WT, greater than 0.94 for TC). The IQR of the distribution is smaller in size, which means that there was good performance across most of the test subjects. Nonetheless, each model generates an outlier subset with Dice values under 0.5 for both TC and ET; subjects with very small or even no enhancing tumor areas will generate extremely low Dice scores, since one small shift would make the Dice value drop to almost zero.

Of particular interest is the attention U-Net TC distribution, whose median is slightly higher but has a similar number of outliers compared to the 3D U-Net model. For the swin U-NETr model, there are more cases under 0.75.

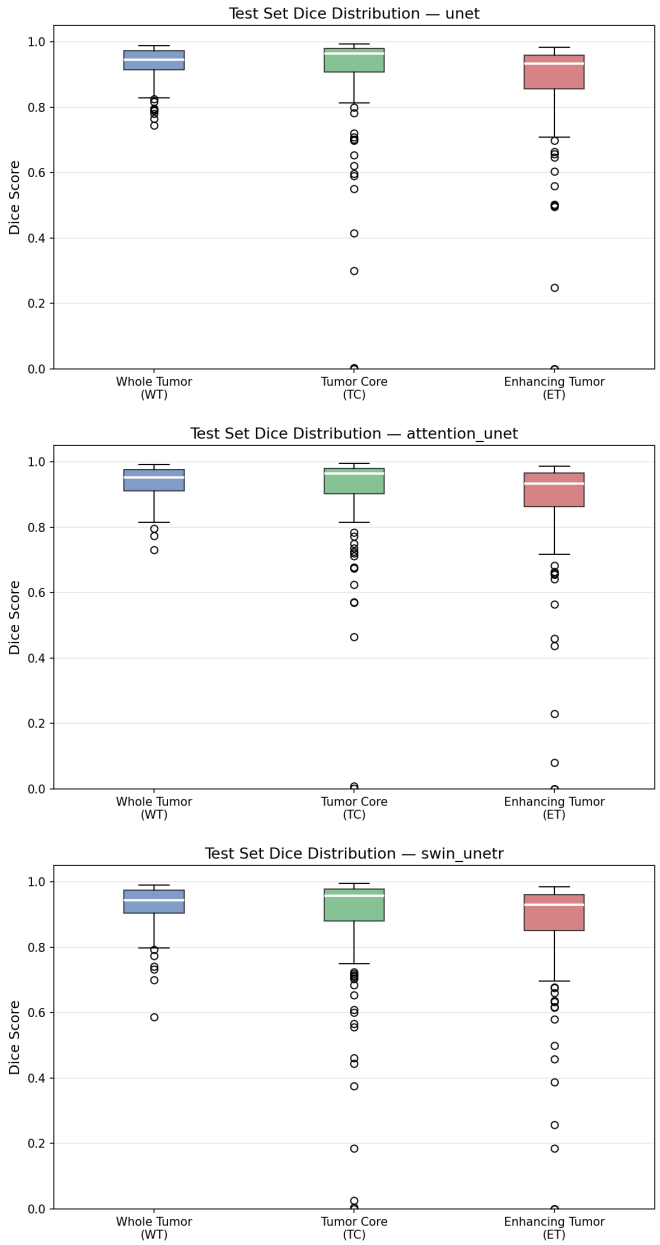

Fig. 2. Boxplots of per-subject Dice scores on the test set for 3D U-Net (top), Attention U-Net (middle), and Swin UNETR (bottom). Each model shows a compact high-performing majority with a long tail of difficult cases, mostly in TC and ET.

4.4 Model Comparison

This can be seen from Fig. 3. From the Dice plot we see that the three networks perform well on WT, and the margin is larger on TC, but the biggest difference is on ET, which is the most difficult sub-region. More specifically, we see that from the HD95 plot that U-Net's ET HD95 has the best performance among all networks and all sub-regions, at a value of 1.78 mm. The HD95 value of TC and ET of Swin UNETR is more than 2.9 mm.

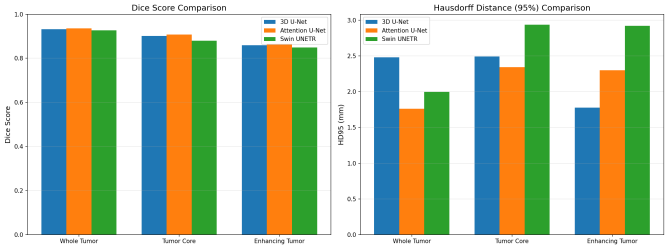

Fig. 3. Side-by-side comparison of Dice scores (left) and HD95 (right) for all three models across WT, TC, and ET. Attention U-Net leads on Dice; 3D U-Net achieves the best HD95 on ET; Swin UNETR lags on both metrics.

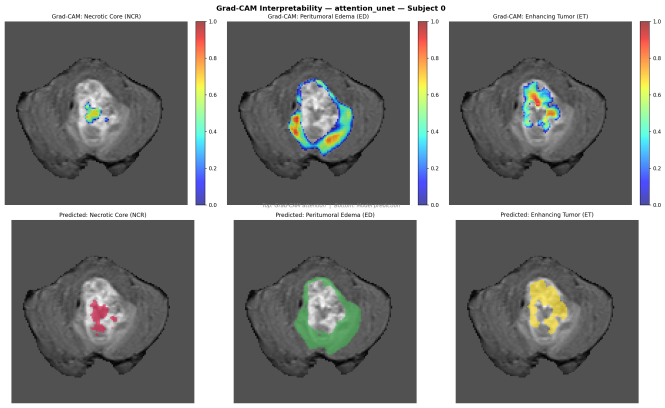

Fig. 4. Grad-CAM activation maps (top row) and model predictions (bottom row) for Attention U-Net on Subject 0. Left to right: Necrotic Core (NCR), Peritumoral Edema (ED), Enhancing Tumor (ET). The peritumoral edema map shows well-concentrated high-activation (red) gradients aligned with the tumor boundary, consistent with anatomically correct segmentation.

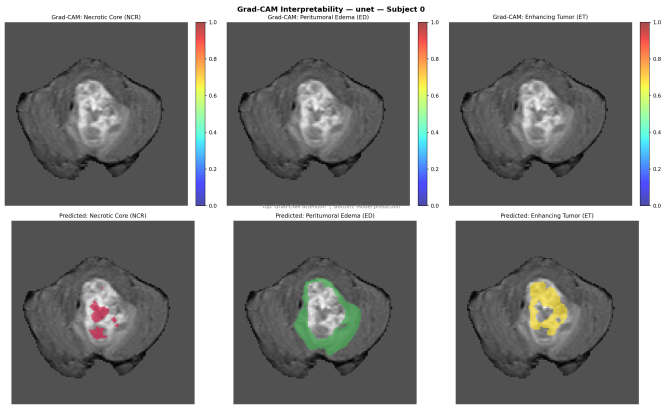

Fig. 5. Grad-CAM for 3D U-Net on Subject 0. Activations are diffuse compared to Attention U-Net, particularly for the necrotic core and enhancing tumor channels, suggesting the model relies on broader contextual regions rather than the precise tumor boundary.

4.5 Grad-CAM Interpretability

Figs. 4–6 show Grad-CAM saliency overlaid on Subject 0's T1ce slice for the three models.

Attention U-Net (Fig. 4) produces the most interpretable activation maps. For the peritumoral edema channel, high-gradient regions (red) are tightly concentrated along the tumor boundary, which aligns with where T2/FLAIR intensity

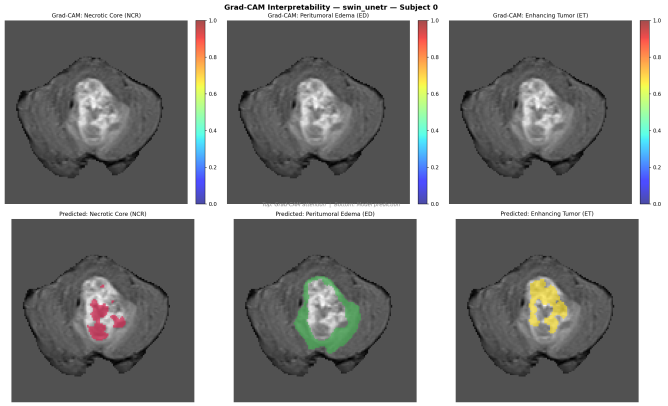

Fig. 6. Grad-CAM for Swin UNETR on Subject 0. The activation maps appear nearly uniform, providing little discriminative spatial information. This suggests that the final feature layer in our Swin UNETR implementation does not localize well, possibly due to incomplete convergence.

transitions occur. For the enhancing tumor, activations localize well to the contrast-enhancing core visible in T1ce. The predicted masks (bottom row) closely match the anatomical structures one would expect, with a compact necrotic core, a well-delineated edema region covering the broader mass, and an enhancing tumor confined to the inner rim.

3D U-Net (Fig. 5) activations are noticeably more diffuse. While the overall predicted masks look reasonable and are spatially similar to Attention U-Net's output, the gradient saliency is spread over a broader region, indicating that the model is not as selective about the specific spatial features it uses. This is exactly what attention gates are designed to address.

Swin UNETR (Fig. 6) shows near-uniform, low-magnitude activations with no clear spatial focus. This is a known limitation of Grad-CAM applied to Transformer-based models, where global self-attention distributes information across many spatial tokens, making local gradient signals less informative. The predictions still look structurally correct (the overall shapes match), but we cannot draw strong conclusions about what the model is actually attending to.

These interpretability results reinforce the quantitative findings: Attention U-Net's spatial gating mechanism not only improves segmentation accuracy but also produces more clinically interpretable feature responses.

5 Discussion

5.1 Why Attention U-Net Performs Best

What makes Attention U-Net outperform the standard 3D U-Net lies clear when theory meets results. At every upsampling stage, attention mechanisms apply selective focus - guided by the data itself - noticed through learned suppression of less relevant activation patterns. Instead of treating all incoming features equally, these gates tune sensitivity based on context. In scans packed with empty space, such precision prevents noise from drowning out subtle but critical signals. Because most volume consists of non-target areas, uncontrolled feature flow risks distorting learning; here, control matters. Small structures gain visibility simply because distraction fades.

Data back this view clearly. Though gains in Dice scores appear small - just 0.004 higher on WT, 0.006 on TC and ET - they show up reliably in each of the three areas tested, appearing again under varied conditions. What stands out more is how much closer the HD95 values get: down by 0.72 mm on WT, 0.15 mm on TC; suggesting the model draws tighter lines around tumors, cutting noise that strays far from actual edges. From Figure 4, Grad-CAM maps reveal why - heat builds along tumor margins in the peritumoral edema layer, exactly at spots where shifts in T2/FLAIR signal offer key hints.

Despite everything, only ET HD95 stands apart - U-Net without attention hits 1.778 mm compared to 2.298 mm with attention. This gap likely stems from excessive signal reduction caused by the attention mechanism, especially within tiny, unstable areas. In cases of minimal ET volume - just a few cubic millimeters - even minor miscalibrations in gate activation may lead the model to overlook scattered enhanced voxels completely, dragging HD95 down sharply.

5.2 Why Swin UNETR Underperforms

What stands out is Swin UNETR falling short of both CNN models - a result that, while clear, fits within expected bounds considering limited training conditions. Behind this gap, several elements probably play a role. First, architectural complexity may slow learning when data stays scarce. Second, reliance on self-attention could hurt performance without enough samples to guide robust feature capture. Third, differences in how positional information gets processed might place it at a disadvantage compared to convolution-based approaches.

One key point: sample efficiency. The self-attention mechanism in Transformers lacks built-in assumptions about space - whereas convolutions treat neighboring voxels as more connected than faraway ones. Because of this difference, Transformers can model a wider range of patterns, yet require larger datasets to pick up basic spatial structures. When trained on around 1,000 examples, the Swin encoder shows signs of incomplete learning, seen in its flattened loss curve near 0.30, while CNNs settle closer to 0.20.

Next, optimization behavior: The same Adam optimizer combined with a cosine decay pattern guided training across all three models. While transformers typically respond well to carefully tuned learning rates, they especially profit when an initial linear ramp spans roughly five to ten cycles prior to decline. Omitting such gradual increase risks erratic shifts during the start, disturbing how attention weights settle.

Third, Grad-CAM shortcomings: Nearly flat activation patterns in Swin UNETR (Fig. 6) arise because Grad-CAM struggles with global attention mechanisms. While convolutional models preserve local structure through spatial feature grids, Transformer architectures let each token interact with all others, spreading influence widely across the input. As a result, gradients lose clear positional meaning, complicating visual interpretation. Such behavior does not indicate incorrect outputs from the model - rather, it hinders how clearly we can trace reasoning paths using heatmaps alone. Though predictions may be accurate, intuitive explanation suffers under these conditions.

5.3 Failure Cases and Hard Sub-Regions

Each model struggles similarly when there is little or even no enhancing tumor present. For certain patients in the BraTS 2021 test data, there are zero label-4 voxels - meaning no active tumor exists. When predictions mark such cases as having ET, errors arise, pushing Dice scores close to zero while inflating HD95 results sharply. That pattern shows clearly in the stretched bottom ends of the ET distribution charts across every method tested. Instead of leaving tiny misclassified blobs unchecked, earlier work suggests cleaning outputs by discarding small clusters below a set size limit - an approach known to reduce average HD95 noticeably. Although simple to implement, we left out this refinement during evaluation.

Occasionally, issues with TC emerge most often in tumors that look atypical - such as infiltrative gliomas lacking a clear dead center, or instances after surgery where original structures are altered. What shows up here isn't merely a flaw in the algorithm, but reflects real uncertainty built into the reference labels. Despite high-quality training data, some boundaries remain unclear by nature, making consistent labeling difficult even for experts. Ambiguity like this sits at the heart of these misclassifications.

5.4 Broader Implications

What stands out most is how small changes in training setup can outweigh flashy architecture choices - especially where data are scarce. Instead of assuming bigger models win, consider that a leaner design might do better; take Attention U-Net, for instance, gaining under half a million weights yet beating Swin UNETR by a wide margin despite its 62-million-parameter bulk. Earlier work lines up with this: Isensee and team showed even basic U-Nets, if tuned right, hold their own against complex custom networks. So when real-world medical projects face tight datasets, placing trust in a solid convolutional backbone enhanced with focused features like attention gates tends to pay off more reliably than betting on oversized Transformers struggling to stabilize during learning.

Still, dismissing Swin UNETR's edge in capability would be premature. With extended training schedules, gradual learning rate increases, or access to more data, performance could exceed that of traditional convolutional networks - just as shown when the model was first introduced. What we present here sticks strictly to experiments run over 100 epochs using one A100 GPU, a setup aligned closely with typical funding limits in university settings.

5.5 Limitations

One key constraint involves the evaluation using just one static data partition; averaging results across several cross-validation splits could strengthen statistical reliability. It is also worth noting that no refinement steps were applied to the initial output of the model - steps previously shown to enhance HD95 scores. Rather than examining many cases, the Grad-CAM interpretation focuses only on a single example case (Subject 0), whereas broader insights might emerge from testing across all subjects while measuring pattern stability. Another aspect left unmeasured includes how long

predictions take and how much memory they consume - factors that often influence real-world medical use.

6 Conclusion

We have presented a close look at three volume-based deep learning designs - 3D U-Net, Attention U-Net, and Swin UNETR - was carried out to assess their ability to segment multiple brain tumor regions within the BraTS 2021 data. Though each model followed the exact same setup, from loss calculation to optimization method, variations in results still emerged after 100 rounds of training. Identical computing resources were used throughout; this removed equipment-related imbalances that might skew outcomes. Data adjustments applied during training stayed consistent across frameworks, helping isolate how structure alone affects accuracy. Because every factor outside network layout remained fixed, observed gaps in output likely stem from design differences instead of external boosts.

Top results appear in Attention U-Unet when tested on all three BraTS areas. With dice outcomes reaching 0.9352 for whole tumor, 0.9077 for tumor core, and 0.8649 for enhancing tumor, performance stands high. Boundary precision leads at 1.762 mm and 2.344 mm for WT and TC respectively - surpassed only within ET among other models. Since the attention gate brings few extra parameters compared to base design, gains remain efficient. Instead of spreading gradients evenly, relevance shifts toward meaningful zones in the scan. Such targeting becomes visible through Grad-CAM maps showing where features matter most.

Despite its simplicity, 3D U-Net proves effective when implemented carefully. With an HD95 of 1.778 mm on ET, it leads in precision for that subregion. Performance consistency stands out - Dice scores show little variation across individuals tested. When speed matters or computing power is constrained, this standard architecture still holds up remarkably well.

Though Swin UNETR carries more parameters and begins with self-supervised pretraining, it scores below the CNN alternatives here. This likely stems from too little optimization time given its size - only 100 epochs - and Transformers' weaker data use per sample. The final loss stays elevated at about 0.30 compared to 0.20 hinting at unused potential. Heatmaps across inputs appear almost identical, further indicating shallow learning. Despite strong starting conditions, progress stalls early.

Down the road, new paths emerge naturally. Training Swin UNETR longer using gradual learning rates along with bigger batches might shrink - or even flip - the current performance difference. Instead of relying only on labeled data, tapping into unused MRI scans from single institutions through semi- or self-supervised methods could strengthen how well both CNNs and Transformers adapt across different scan settings. When assessing results, basic cleanup techniques - like weeding out false-positive enhancing tumor areas by grouping nearby regions - could lift HD95 scores across the board without touching model weights. Confidence matters just as much as correctness in real medical use; adding ways to measure prediction certainty - say, via repeated dropout runs or multiple model outputs - moves these systems one step closer to actual clinics.

This study offers a clear, repeatable analysis to help choose models for brain tumor segmentation where resources are limited. While focused on practical needs, it supports decision-making in academic and medical research contexts. By emphasizing openness and consistency, the approach allows others to follow and verify results. Its value lies in balancing rigor with accessibility for real-world applications.

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